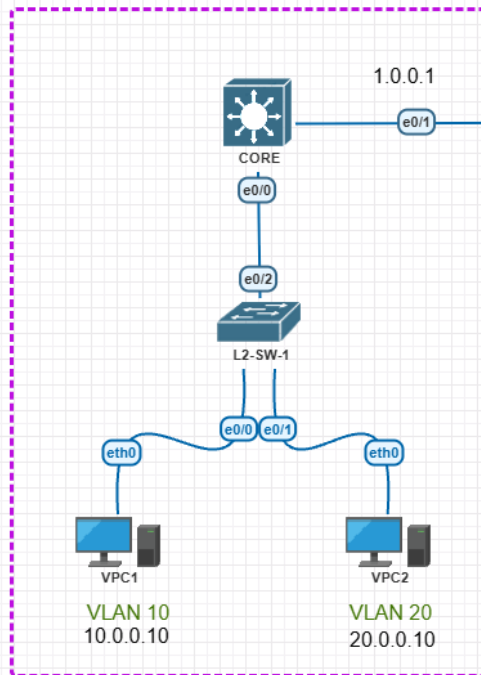
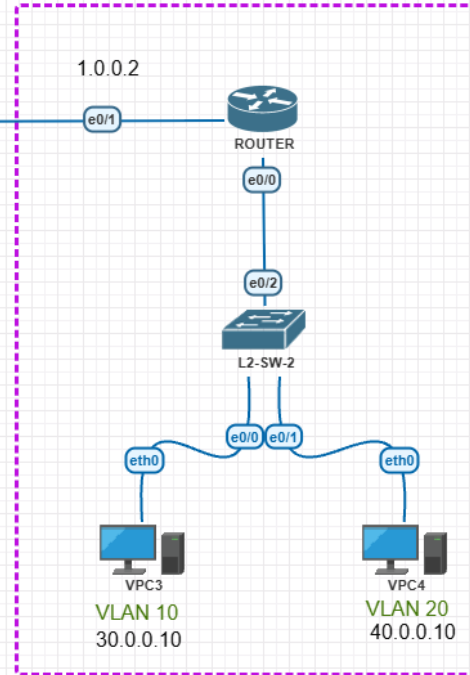




SWITCH VIRTUAL INTERFACE



ROUTER ON-A-STICK

 L2-SW-1

```
Switch>enable
Switch#conf t
Enter configuration commands, one per line.  End with CNTL/Z.
Switch(config)#vlan 10
Switch(config-vlan)#name hr
Switch(config-vlan)#exit
Switch(config)#vlan 20
Switch(config-vlan)#name sales
Switch(config-vlan)#exit
Switch(config)#int e0/0
Switch(config-if)#switchport access vlan 10
Switch(config-if)#exit
Switch(config)#int e0/1
Switch(config-if)#switchport access vlan 20
Switch(config-if)#exit
Switch(config)#int e0/2
Switch(config-if)#switchport trunk encapsulation
*Sep 18 12:32:26.625: %PNP-6-PNP_DISCOVERY_STOPPED: PnP Discovery stopped (Config Wizard)
Switch(config-if)#switchport trunk encapsulation dot1q
Switch(config-if)#switchport mode trunk
Switch(config-if)#exit
Switch(config)#
```

```
Switch>enable
Switch#conf t
Enter configuration commands, one per line.  End with CNTL/Z.
Switch(config)#vlan 10
Switch(config-vlan)#exit
Switch(config)#vlan 20
Switch(config-vlan)#exit
Switch(config)#interface vlan 10
Switch(config-if)#
*Sep 18 12:38:09.277: %LINEPROTO-5-UPDOWN: Line protocol on Interface Vlan10, changed state to down
Switch(config-if)#ip add
*Sep 18 12:38:53.141: %PNP-6-PNP_DISCOVERY_STOPPED: PnP Discovery stopped (Config Wizard)
Switch(config-if)#ip add 10.0.0.1 255.0.0.0
Switch(config-if)#exit
Switch(config)#interface vlan 20
Switch(config-if)#ip add
*Sep 18 12:40:36.220: %LINEPROTO-5-UPDOWN: Line protocol on Interface Vlan20, changed state to down
Switch(config-if)#ip add 20.0.0.1 255.0.0.0
Switch(config-if)#exit
Switch(config)#ip routing
Switch(config)#exit
```

```
Switch#conf t
Enter configuration commands, one per line. End with CNTL/Z.
Switch(config)#ip routing
Switch(config)#do sh ip int br
Interface                IP-Address      OK? Method Status          Protocol
Ethernet0/0              unassigned      YES unset  up              up
Ethernet0/1              unassigned      YES unset  up              up
Ethernet0/2              unassigned      YES unset  down            down
Ethernet0/3              unassigned      YES unset  down            down
Vlan10                   10.0.0.1        YES manual administratively down down
Vlan20                   20.0.0.1        YES manual administratively down down
Switch(config)#int e0/1
Switch(config-if)#no switchport
Switch(config-if)#ip add
*Sep 18 13:10:27.493: %LINK-3-UPDOWN: Interface Ethernet0/1, changed state to up
*Sep 18 13:10:28.493: %LINEPROTO-5-UPDOWN: Line protocol on Interface Ethernet0/1, changed state to up
Switch(config-if)#ip add 1.0.0.1 255.0.0.0
Switch(config-if)#no sh
Switch(config-if)#exit
Switch(config)#do sh ip int br
Interface                IP-Address      OK? Method Status          Protocol
Ethernet0/0              unassigned      YES unset  up              up
Ethernet0/1              1.0.0.1         YES manual up              up
Ethernet0/2              unassigned      YES unset  down            down
Ethernet0/3              unassigned      YES unset  down            down
Vlan10                   10.0.0.1        YES manual administratively down down
Vlan20                   20.0.0.1        YES manual administratively down down
Switch(config)#ip route 0.0.0.0 0.0.0.0 1.0.0.2
Switch(config)#exit
```

ROUTER

```
3, changed state to down
ROUTER(config)#int e0/0
ROUTER(config-if)#no sh
ROUTER(config-if)#exit
ROUTER(config)#int e0/0.1
ROUTER(config-subif)#encapsulation dot1q 10
ROUTER(config-subif)#ip add 30.0.0.1 255.0.0.0
ROUTER(config-subif)#no sh
ROUTER(config-subif)#exit
ROUTER(config)#int e0/0.2
ROUTER(config-subif)#encapsulation dot1q 20
ROUTER(config-subif)#ip add 40.0.0.1 255.0.0.0
ROUTER(config-subif)#no sh
ROUTER(config-subif)#exit
ROUTER(config)#int e0/1
ROUTER(config-if)#ip add 1.0.0.2 255.0.0.0
ROUTER(config-if)#no sh
ROUTER(config-if)#exit
ROUTER(config)#ip route 0.0.0.0
*Oct  3 08:24:35.738: %PNP-6-PNP_DISCOVERY_STOPPED: PnP Discovery stopped (Config Wizard)
ROUTER(config)#ip route 0.0.0.0 0.0.0.0 1.0.0.1
ROUTER(config)#exit
```

```
ROUTER#sh ip ro
```

```
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2
       i - IS-IS, su - IS-IS summary, L1 - IS-IS level-1, L2 - IS-IS level-2
       ia - IS-IS inter area, * - candidate default, U - per-user static route
       o - ODR, P - periodic downloaded static route, H - NHRP, l - LISP
       a - application route
       + - replicated route, % - next hop override, p - overrides from PfR
```

```
Gateway of last resort is 1.0.0.1 to network 0.0.0.0
```

```
S*    0.0.0.0/0 [1/0] via 1.0.0.1
      1.0.0.0/8 is variably subnetted, 2 subnets, 2 masks
C      1.0.0.0/8 is directly connected, Ethernet0/1
L      1.0.0.2/32 is directly connected, Ethernet0/1
      30.0.0.0/8 is variably subnetted, 2 subnets, 2 masks
C      30.0.0.0/8 is directly connected, Ethernet0/0.1
L      30.0.0.1/32 is directly connected, Ethernet0/0.1
      40.0.0.0/8 is variably subnetted, 2 subnets, 2 masks
C      40.0.0.0/8 is directly connected, Ethernet0/0.2
L      40.0.0.1/32 is directly connected, Ethernet0/0.2
```

```
ROUTER#sh ip int br
```

Interface	IP-Address	OK?	Method	Status	Prot
ocol					
Ethernet0/0	unassigned	YES	TFTP	up	up
Ethernet0/0.1	30.0.0.1	YES	manual	up	up
Ethernet0/0.2	40.0.0.1	YES	manual	up	up
Ethernet0/1	1.0.0.2	YES	manual	up	up
Ethernet0/2	unassigned	YES	TFTP	down	down
Ethernet0/3	unassigned	YES	TFTP	down	down

```
VPC1
PC1> ping 40.0.0.10

4 bytes from 40.0.0.10 icmp_seq=1 ttl=62 time=2.499 ms
4 bytes from 40.0.0.10 icmp_seq=2 ttl=62 time=1.662 ms
4 bytes from 40.0.0.10 icmp_seq=3 ttl=62 time=1.636 ms
4 bytes from 40.0.0.10 icmp_seq=4 ttl=62 time=1.963 ms
4 bytes from 40.0.0.10 icmp_seq=5 ttl=62 time=2.496 ms
```

```
VPC2
VPC2> ping 30.0.0.10

84 bytes from 30.0.0.10 icmp_seq=1 ttl=62 time=1.649 ms
84 bytes from 30.0.0.10 icmp_seq=2 ttl=62 time=1.860 ms
84 bytes from 30.0.0.10 icmp_seq=3 ttl=62 time=1.820 ms
84 bytes from 30.0.0.10 icmp_seq=4 ttl=62 time=1.842 ms
84 bytes from 30.0.0.10 icmp_seq=5 ttl=62 time=1.807 ms
```